

لتحويل العدد المركب من الصيغة الجبرية $x+iy$ الى الصورة القطبية $r e^{i\theta}$
 $r(\cos(\theta) + i \sin(\theta))$

نتبع ما يلي
 (1) نوجد المقيا (r) او يرمز له بالرمز $|z|$ او $\text{mod}(z)$

من العلاقة

$$r = \sqrt{x^2 + y^2}$$

$$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$\tan^{-1}(1) = \frac{\pi}{4}$$

$$\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

(argument) $\arg(z)$ نوجد له

لايجاد له أولاً نحدد موقع العدد المركب

(1) اذا كان يقع في الربع الاول (x, y) موجب

$$\arg(z) = \tan^{-1}\left|\frac{y}{x}\right|$$

(2) اذا كان يقع في الربع الثاني $(x \text{ سالب}, y \text{ موجب})$

$$\arg(z) = \pi - \tan^{-1}\left|\frac{y}{x}\right|$$

(3) اذا كان يقع في الربع الثالث $(x \text{ سالب}, y \text{ سالب})$

$$\arg(z) = \pi + \tan^{-1}\left|\frac{y}{x}\right|$$

وإذا نريد الزاوية بالقياس نستخدم

$$\arg(z) = \pi + \tan^{-1}\left|\frac{y}{x}\right|$$

(4) اذا كان يقع في الربع الرابع $(x \text{ موجب}, y \text{ سالب})$

$$\arg(z) = -\tan^{-1}\left|\frac{y}{x}\right|$$

النتيجة المختصرة ما يستخدم

الاول	الثاني
$\tan^{-1}\left \frac{y}{x}\right $	$\pi - \tan^{-1}\left \frac{y}{x}\right $
$-\tan^{-1}\left \frac{y}{x}\right $	$\pi + \tan^{-1}\left \frac{y}{x}\right $
الرابع	الثالث
$-\tan^{-1}\left \frac{y}{x}\right $	$\pi + \tan^{-1}\left \frac{y}{x}\right $

$$-\pi + \tan^{-1}\left|\frac{y}{x}\right|$$

write the following numbers in polar form
المسوحة ضوئياً بـ

① $z = 1 + i$

Sol

اولاً نوجد مقدارها

$$\therefore x = 1, y = 1$$

$$r = \sqrt{x^2 + y^2} = \sqrt{(1)^2 + (1)^2} = \sqrt{1+1} = \sqrt{2}$$

$$\therefore \text{mod}(z) = r = \boxed{\sqrt{2}}$$

العدد عند زاوية اول

$$\arg(z) = \tan^{-1} \left(\frac{y}{x} \right)$$

$$= \tan^{-1} \left(\frac{1}{1} \right) = \tan^{-1}(1) = \frac{\pi}{4}$$

$$\therefore \arg(z) = \frac{\pi}{4}$$

① in polar form

$$\therefore r = \sqrt{2}, \theta = \frac{\pi}{4}$$

$$r (\cos(\theta) + i \sin(\theta))$$

$$\sqrt{2} \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right)$$

$$(2) -\sqrt{3} + i$$

$$x = -\sqrt{3}, y = 1$$

او لا توجد
المقياس
واسعة

$$(1) \therefore r = \sqrt{x^2 + y^2} = \sqrt{(-\sqrt{3})^2 + (1)^2} = \sqrt{3+1} = \sqrt{4} = 2$$

$$r = 2$$

(2) العدد في الربع الثاني لأن x سالب، y موجب

$$\theta = \arg(z) = \pi - \tan^{-1} \left| \frac{1}{\sqrt{3}} \right|$$

$$\theta = \pi - \frac{\pi}{6} = \frac{6\pi - \pi}{6}$$

$$\theta = \frac{5\pi}{6}$$

$$\begin{aligned} \text{نعلم أنه} \\ \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \\ = \frac{\pi}{6} \end{aligned}$$

نكتب x
لا بد من إشارات
لأنه بالقانون
قيمة مطلقة

$$(1) \text{ in polar form } r (\cos(\theta) + i \sin(\theta))$$

$$\text{الصورة القطبية} \quad z \left(\cos \left(\frac{5\pi}{6} \right) + i \sin \frac{5\pi}{6} \right)$$

$$(3) z = \frac{1}{2} - \frac{\sqrt{3}}{2} i$$

$$x = \frac{1}{2}, y = -\frac{\sqrt{3}}{2}$$

$$r = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{3}{4}} = \sqrt{\frac{4}{4}} = \sqrt{1} = 1 \Rightarrow r = 1$$

$$\theta = -\tan^{-1} \left| \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} \right| = -\tan^{-1}(\sqrt{3})$$

(1) polar form

$$1 \left(\cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right) \right) = \cos \frac{\pi}{3} - i \sin \frac{\pi}{3}$$

$$\theta = -\frac{\pi}{3}$$

$$\begin{aligned} \text{ملاحظة} \\ \sin(-\theta) \\ = -\sin(\theta) \\ \cos(-\theta) \\ = \cos(\theta) \end{aligned}$$

Express $-3 + 3\sqrt{3}i$

(1) Express in polar
الشكل

$$x = -3 \quad y = 3\sqrt{3}$$

$$r = \sqrt{x^2 + y^2} = \sqrt{(-3)^2 + (3\sqrt{3})^2}$$

$$= \sqrt{9 + (9)(3)}$$

$$= \sqrt{9 + 27} = \sqrt{36} = 6$$

$$r = 6$$

$$\theta = \pi - \tan^{-1} \left| \frac{3\sqrt{3}}{-3} \right|$$

$$\theta = \pi - \tan^{-1}(\sqrt{3})$$

$$\theta = \pi - \frac{\pi}{3}$$

$$\theta = \frac{3\pi - \pi}{3} = \frac{2\pi}{3}$$

$$\theta = \frac{2\pi}{3}$$

$$z = \left[6, \frac{2\pi}{3} \right]$$

$$= 6 \left(\cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) \right)$$



بكل ثور غالي

$$\begin{aligned} \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) &= \frac{\pi}{6} \\ \tan^{-1}(\sqrt{3}) &= \frac{\pi}{3} \\ \tan^{-1}(1) &= \frac{\pi}{4} \end{aligned}$$

(2)

=

Express

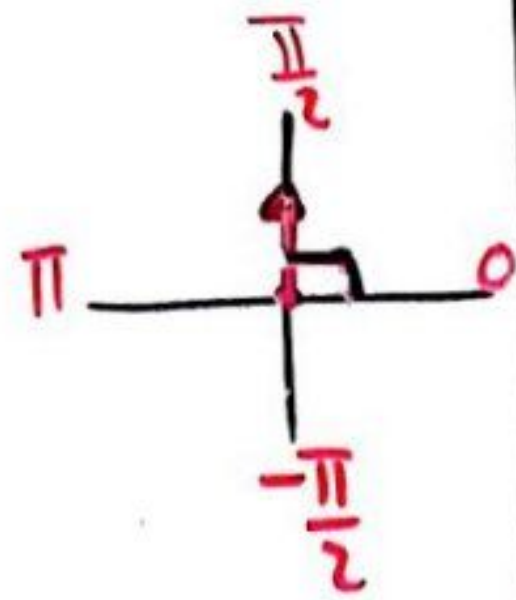
i

① Express in polar
القطبية

$$x = 0$$

$$y = 1$$

$$r = 1$$



$$\theta = \arg(z) = \frac{\pi}{2}$$

$$z = \left[1, \frac{\pi}{2} \right]$$

$$= 1 \left(\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \right)$$

$$\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right)$$

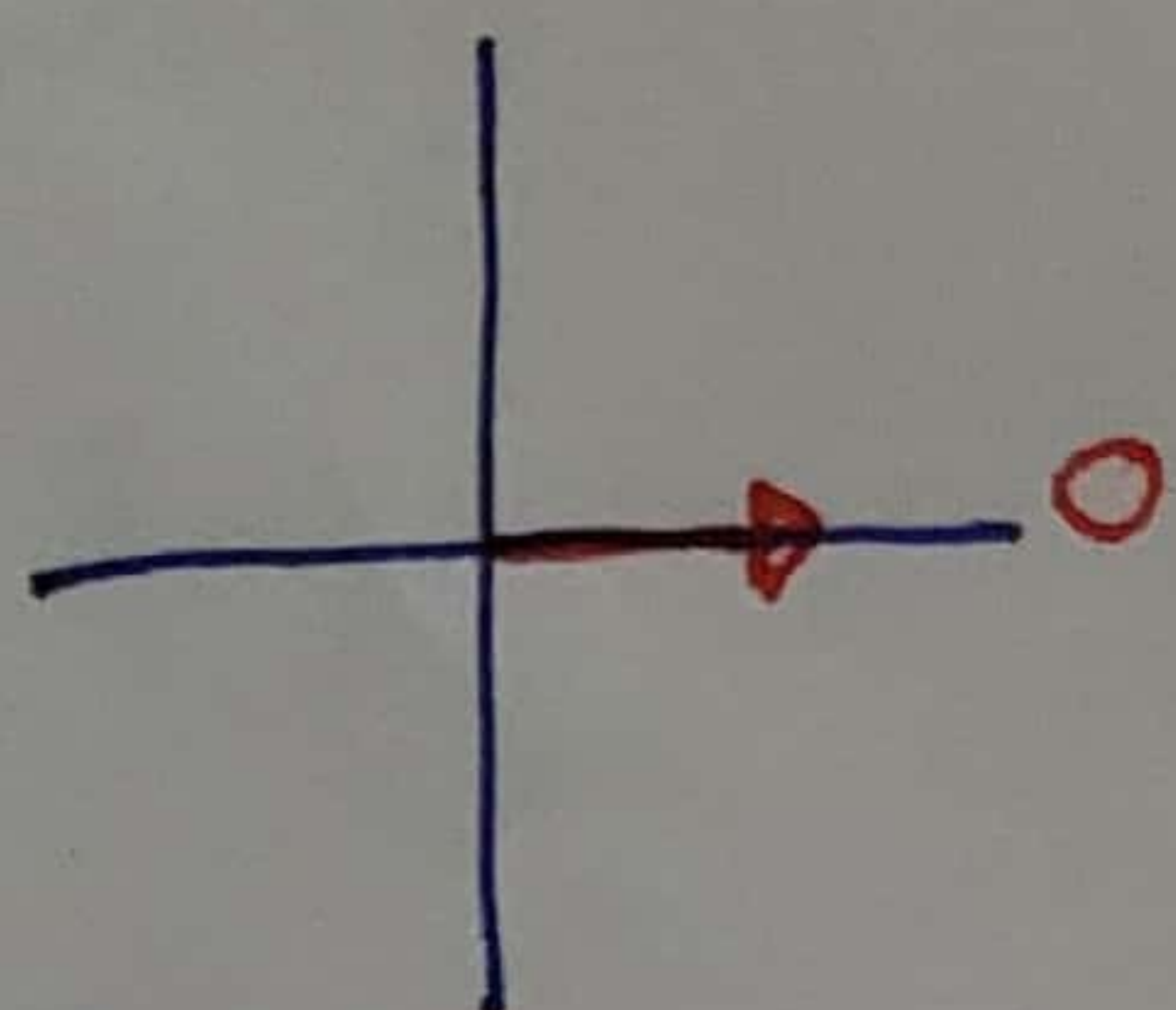
لکھتور غالی

$$\begin{aligned} \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) &= \frac{\pi}{6} \\ \tan^{-1}(\sqrt{3}) &= \frac{\pi}{3} \\ \tan^{-1}(1) &= \frac{\pi}{4} \end{aligned}$$

① if $z = a$, $a > 0$

then $\text{mod}(z) = a$, $\arg(z) = 0$

المقياس r الزاوية

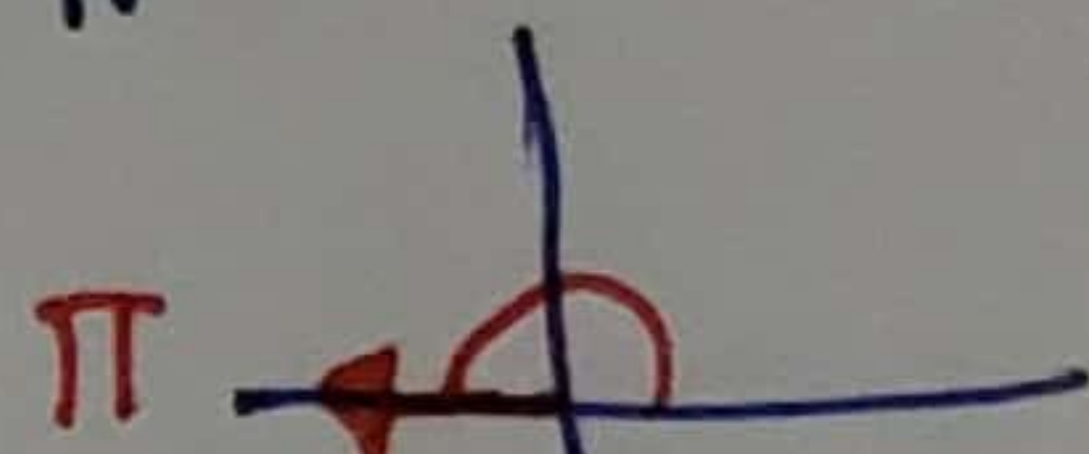


② $z = -a$, $a > 0$

حقيقي صرفاً سالب

then $\text{mod}(z) = a$, $\arg(z) = \pi$

المقياس r الزاوية



المقياس دائماً موجب

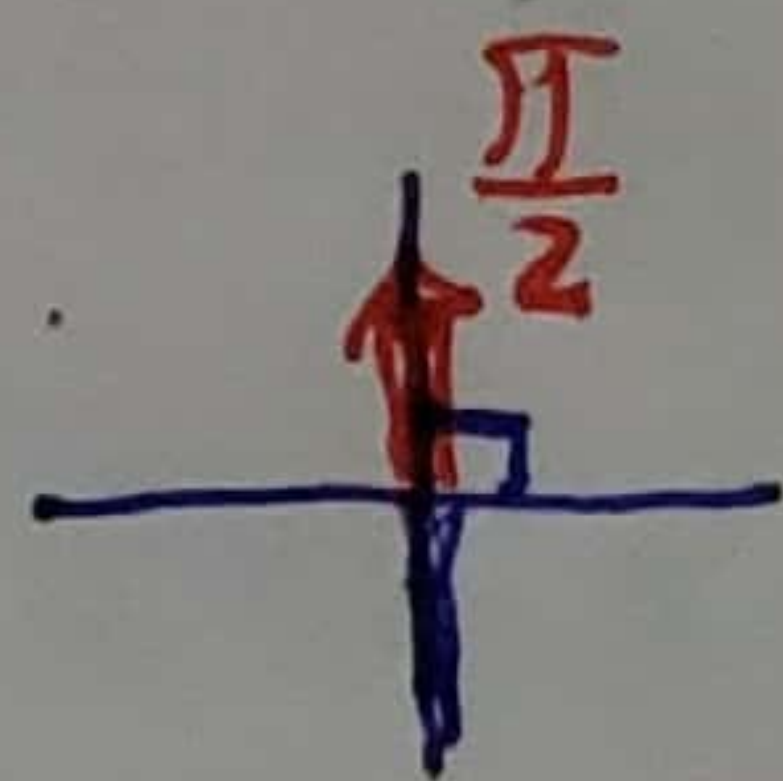
Polar form of z is $a(\cos(\pi) + i\sin(\pi))$

③ $z = ai$, $a > 0$

تخيلي صرفاً موجب

then $\text{mod}(z) = a$, $\arg(z) = \frac{\pi}{2}$

معامل i بال موجب الزاوية



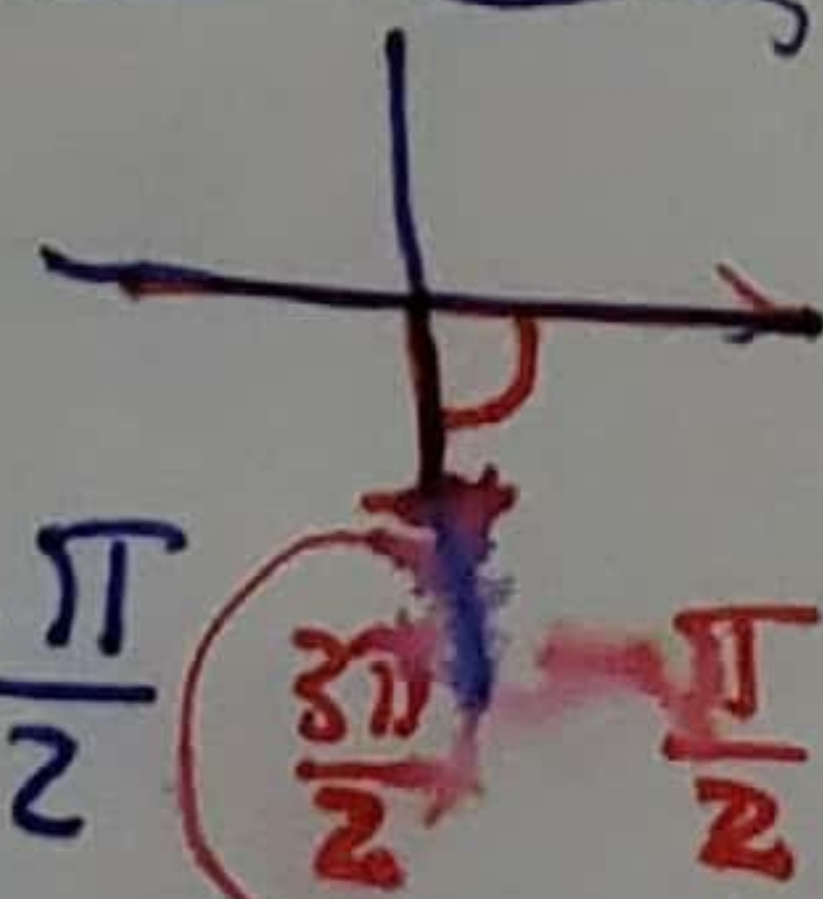
Polar form z is $a(\cos(\frac{\pi}{2}) + i\sin(\frac{\pi}{2}))$

④ $z = -ai$, $a > 0$

تخيلي صرفاً سالب

$\text{mod}(z) = a$

$\arg(z) = -\frac{\pi}{2}$ or $\frac{3\pi}{2}$



معامل i بال موجب

polar form of z

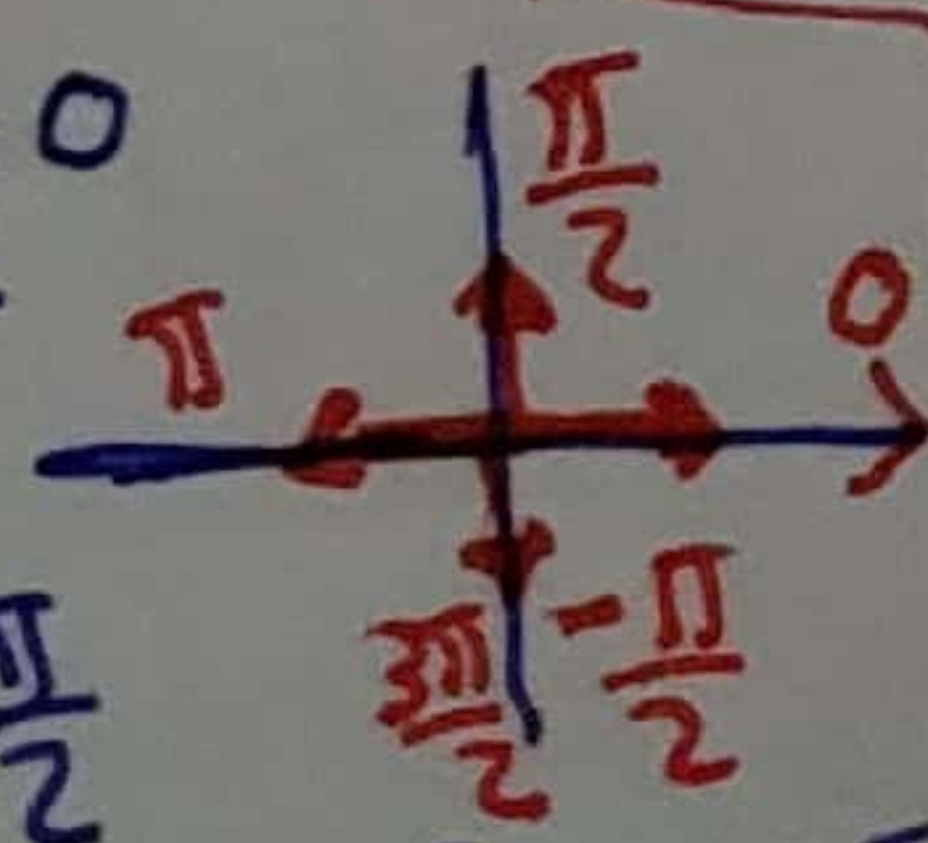
is $a(\cos(\frac{\pi}{2}) - i\sin(\frac{\pi}{2}))$

$z = 3 \rightarrow r = 3, \theta = 0$

$z = -3 \rightarrow r = 3, \theta = \pi$

$z = 3i \rightarrow r = 3, \theta = \frac{\pi}{2}$

$z = -3i \rightarrow r = 3, \theta = -\frac{\pi}{2}$ or $\frac{3\pi}{2}$



$z = 1, r = 1, \theta = 0$

$z = -1, r = 1, \theta = \pi$

$z = i, r = 1, \theta = \frac{\pi}{2}$

$z = -i, r = 1, \theta = -\frac{\pi}{2}$ or $\frac{3\pi}{2}$

$z = -i \rightarrow r = 1, \theta = -\frac{\pi}{2}$

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المثال

القطبي Polar form

$$① z = 5 \rightarrow z = 5 (\cos(0) + i \sin(0))$$

$$② z = -3 \rightarrow z = 3 (\cos(\pi) + i \sin(\pi))$$

$$③ z = 2i \rightarrow z = 2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)$$

$$④ z = -2i \rightarrow z = 2 \left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right)$$

or

$$z = 2 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right)$$

$$z = i = 1 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$$

$$z = -i = 1 \left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right) = \cos \frac{\pi}{2} - i \sin \frac{\pi}{2}$$

$$z = -1 = 1 (\cos(\pi) + i \sin \pi) = \cos \pi + i \sin \pi$$

$$z = \frac{1+i}{1-i} \rightarrow z = i \rightarrow z = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$$

$$z = \frac{1-i}{1+i} \rightarrow z = -i \rightarrow z = \cos \left(\frac{\pi}{2} \right) - i \sin \frac{\pi}{2}$$

$$z = \left(\frac{1+i}{1-i} \right)^4 \rightarrow z = i^4 = 1 \rightarrow z = \cos(0) + i \sin(0)$$

$$z = \frac{1+2i}{1-(1-i)^2}$$

$$(1-i)^2 = -2i$$

$$\text{then } z = \frac{1+2i}{1-(-2i)} = \frac{1+2i}{1+2i} = 1$$

$$z = \cos(0) + i \sin(0)$$

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Express $\frac{1+i}{1-i}$

① Express in polar
نسبة القطبية

$$\frac{1+i}{1-i} = \frac{1+i}{1-i} \cdot \frac{1+i}{1+i}$$

$$= \frac{(1+i)^2}{(1)^2 + (1)^2}$$

$$= \frac{2i}{2} = i$$

$$r = 1$$

$$x = 0, y = 1$$

للسرعة

$$(1+i)^2 = 2i$$

$$(1-i)^2 = -2i$$

$$(1)^2 + (1)^2 = 1 + 1 = 2$$

كل ثور

$$\theta = \frac{\pi}{2}$$

$$i = \cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)$$



(7)

π

i

Express

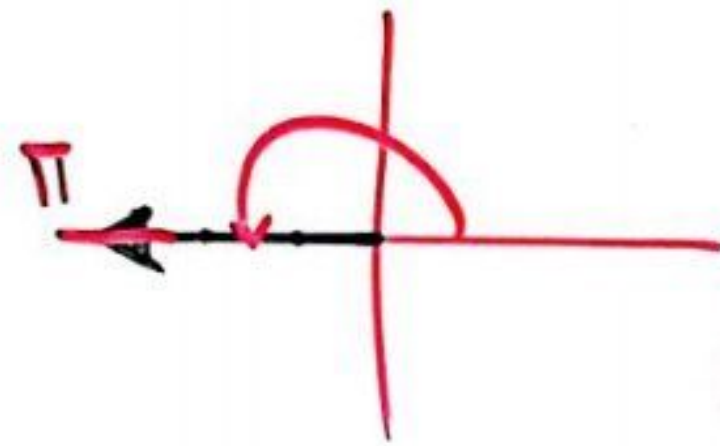
-3

① Express in polar
القطبية

$$x = -3$$

$$y = 0$$

$$r = 3$$



$$\theta = \arg(z) = \pi$$

$$z = [3, \pi]$$

$$= 3 (\cos(\pi) + i \sin(\pi))$$

②

بكل ثور غالي

$$\begin{aligned} \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) &= \frac{\pi}{6} \\ \tan^{-1}(\sqrt{3}) &= \frac{\pi}{3} \\ \tan^{-1}(1) &= \frac{\pi}{4} \end{aligned}$$

Express $\frac{1+2i}{1-(1-i)^2}$

① Express in polar
القطبية

$$\frac{1+2i}{1-(1-i)^2}$$

$$= \frac{1+2i}{1-(-2i)} = \frac{1+2i}{1+2i} = 1$$

$$x=1, y=0$$

$$r=1$$

$$\theta=0$$

$$1 = (\cos(0) + i \sin(0))$$

للسرعة

$$(1+i)^2 = 2i$$

$$(1-i)^2 = -2i$$

$$(1)^2 + (1)^2 = 1+1=2$$

كل ثور

②

Express $\frac{2+6\sqrt{3}i}{5+\sqrt{3}i}$

① Express in polar
 القطبية

نسبة

$$\begin{aligned} \frac{2+6\sqrt{3}i}{5+\sqrt{3}i} &= \frac{2+6\sqrt{3}i}{5+\sqrt{3}i} \cdot \frac{5-\sqrt{3}i}{5-\sqrt{3}i} \\ &= \frac{10 - 2\sqrt{3}i + 30\sqrt{3}i - (6)(3)i}{(5)^2 + (\sqrt{3})^2} \\ &= \frac{10 - 2\sqrt{3}i + 30\sqrt{3}i + 18}{25 + 3} \\ &= \frac{28 + 28\sqrt{3}i}{28} = \frac{28}{28} + \frac{28\sqrt{3}i}{28} = 1 + \sqrt{3}i \end{aligned}$$

لج

② EXP

$x=1, y=\sqrt{3}$

$r = \sqrt{(1)^2 + (\sqrt{3})^2} = \sqrt{1+3}$

$= \sqrt{4} = 2$

$r=2$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\theta = \frac{\pi}{3}$

$1+\sqrt{3}i = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$= 1 + \sqrt{3}i$

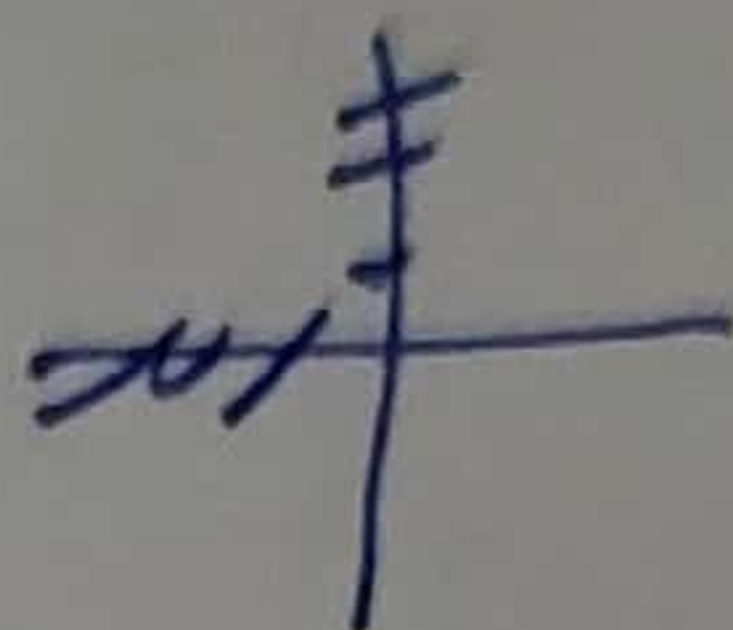
$$④ \quad z = -3 + 4i$$

$$x = -3, \quad y = 4$$

$$r = \sqrt{(-3)^2 + (4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\boxed{r = 5}$$

$$\theta = \pi - \tan^{-1}\left(\frac{4}{3}\right)$$



غير متغير تبين كما هو

$$\theta = \pi - \tan^{-1}\left(\frac{4}{3}\right)$$

in polar form

$$5 \left(\cos \left(\pi - \tan^{-1}\left(\frac{4}{3}\right) \right) + i \sin \left(\pi - \tan^{-1}\left(\frac{4}{3}\right) \right) \right)$$

$$r = \frac{\sqrt{2}}{2}, \quad \theta = \frac{\pi}{4}$$

$$\frac{\sqrt{2}}{2} \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right)$$

$$\frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \text{ ملاحظة}$$

$$⑤ \quad z = \frac{1}{1-i}$$

نضرب بالمرافق

$$z = \frac{1}{1-i} \cdot \frac{1+i}{1+i} = \frac{1+i}{(1)^2 + (1)^2}$$

$$z = \frac{1+i}{2} = \frac{1}{2} + \frac{1}{2}i$$

إذا كان الجذر حقيقي = الجذر التجديدي

وكان العدد في الربع الأول

فإنه لمعة $\frac{\pi}{4}$

$$x = \frac{1}{2}, \quad y = \frac{1}{2} \Rightarrow r = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2}$$

$$\theta = \frac{\pi}{4}$$

$$r = \frac{\sqrt{2}}{2}$$

$$= \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{2}{4}} = \frac{\sqrt{2}}{2}$$

If $z = -3 - 3\sqrt{3}i$ then in polar form of z is...

sol

$$x = -3 \quad y = -3\sqrt{3}$$

$$r = \sqrt{(-3)^2 + (-3\sqrt{3})^2} = \sqrt{9 + 9(3)} = \sqrt{9 + 27} \\ = \sqrt{36} = 6 \Rightarrow \boxed{r=6}$$

$$\therefore \theta = \pi + \tan^{-1} \left| \frac{3\sqrt{3}}{3} \right|$$

$$\theta = \pi + \tan^{-1}(\sqrt{3}) \\ = \pi + \frac{\pi}{3} = \frac{4\pi}{3}$$

$$\theta = \frac{4\pi}{3}$$

$$\therefore z = 6 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right)$$

نلاحظ أنه هذا الجواب غير موجود في الاختيارات ولعدد طرق

في إيجاد القياس السالب للزاوية

في المثال
نلاحظ أن $\frac{4\pi}{3}$ وذلك فنقص
هنا 2π

$$\frac{4\pi}{3} - 2\pi = \frac{4\pi - 6\pi}{3} = -\frac{2\pi}{3}$$

$$\therefore \theta = -\frac{2\pi}{3}$$

$$6 \left(\cos \left(-\frac{2\pi}{3} \right) + i \sin \left(-\frac{2\pi}{3} \right) \right) = 6 \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right)$$

وهو الجواب الأول

الخيارات

$$6 \left(\cos \left(\frac{2\pi}{3} \right) - i \sin \left(\frac{2\pi}{3} \right) \right)$$

$$6 \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right)$$

$$6 \left(\cos \left(\frac{\pi}{3} \right) + i \sin \frac{\pi}{3} \right)$$

$$6 \left(\cos \left(\frac{\pi}{6} \right) + i \sin \frac{\pi}{6} \right)$$

تذكر أن

$$\cos(-\theta) \\ = \cos(\theta)$$

$$\sin(-\theta) \\ = -\sin(\theta)$$

If $z = \frac{2-2i}{4i}$ then in polar form of z is --

$$z = \frac{2-2i}{4i} \cdot \frac{-4i}{-4i} = \frac{-8i+8i^2}{(0)^2+(4)^2}$$

$$= \frac{-8i-8}{16} = -\frac{8}{16} - \frac{8i}{16} = -\frac{1}{2} - \frac{1}{2}i$$

$$x = -\frac{1}{2}, \quad y = -\frac{1}{2}$$

$$r = \sqrt{\left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{2}{4}} = \sqrt{\frac{1}{2}}$$

$$r = \frac{1}{\sqrt{2}}$$

$$\theta = \pi + \tan^{-1} \left| \frac{-\frac{1}{2}}{-\frac{1}{2}} \right|$$

$$= \pi + \tan^{-1}(1) = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$\frac{1}{\sqrt{2}} \left(\cos\left(\frac{5\pi}{4}\right) + i \sin\left(\frac{5\pi}{4}\right) \right)$$

الحل

$$2 \left(\cos\left(\frac{5\pi}{4}\right) + i \sin\left(\frac{5\pi}{4}\right) \right)$$

$$\frac{1}{\sqrt{2}} \left(\cos\left(\frac{5\pi}{4}\right) + i \sin\left(\frac{5\pi}{4}\right) \right)$$

$$2 \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right)$$

none of these

تبين آخر العدد

$$\frac{2-2i}{4i} = \frac{2}{4i} - \frac{2i}{4i}$$

$$= \frac{1}{2i} - \frac{1}{2}$$

$$= -\frac{1}{2}i - \frac{1}{2}$$

If $z = \frac{1-\sqrt{3}i}{8}$ then in polar form of z is --

$$z = \frac{1}{8} - \frac{\sqrt{3}}{8}i$$

$$x = \frac{1}{8}, \quad y = -\frac{\sqrt{3}}{8}$$

$$r = \sqrt{\left(\frac{1}{8}\right)^2 + \left(\frac{\sqrt{3}}{8}\right)^2} = \sqrt{\frac{1}{64} + \frac{3}{64}} = \sqrt{\frac{4}{64}} = \frac{2}{8}$$

$$r = \frac{1}{4}$$

$$\theta = \tan^{-1} \left(\frac{-\frac{\sqrt{3}}{8}}{\frac{1}{8}} \right) = -\tan^{-1}(\sqrt{3})$$

$$\theta = -\frac{\pi}{3}$$

الاجابة

$$\frac{1}{4} \left(\cos\left(\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{3}\right) \right)$$

$$\frac{1}{4} \left(\cos\left(\frac{5\pi}{4}\right) + i \sin\left(\frac{5\pi}{4}\right) \right)$$

$$\frac{1}{4} \left(\cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right) \right)$$

$$\frac{1}{4} \left(\cos\left(\frac{\pi}{3}\right) - i \sin\left(\frac{\pi}{3}\right) \right)$$

$$\frac{1}{4} \left(\cos\left(\frac{\pi}{3}\right) - i \sin\left(\frac{\pi}{3}\right) \right)$$

the polar form of ...

$$\textcircled{1} \quad \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$\textcircled{2} \quad -\sqrt{3} + i$$

$$\textcircled{3} \quad -1 - i$$

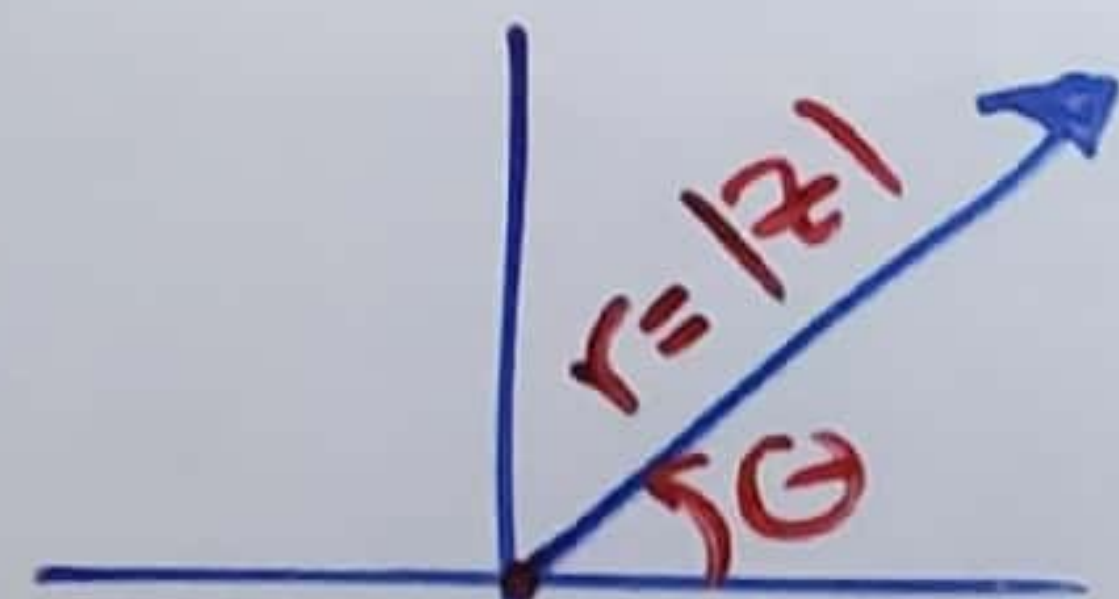
$$\textcircled{4} \quad \frac{1-i}{1+i}$$

$$\textcircled{5} \quad \frac{1+7i}{(2-i)^2}$$

$$\textcircled{6} \quad \frac{2+6\sqrt{3}i}{5+\sqrt{3}i}$$

$$z = x + yi$$

$$(x, y)$$



r or |z|

$$\text{mod}(z) = \sqrt{x^2 + y^2}$$

$$\pi - \tan^{-1}\left|\frac{y}{x}\right| \quad \tan^{-1}\left|\frac{y}{x}\right|$$

$$\pi + \tan^{-1}\left|\frac{y}{x}\right| \quad -\tan^{-1}\left|\frac{y}{x}\right|$$

$$-\pi + \tan^{-1}\left|\frac{y}{x}\right|$$

$$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$\tan^{-1}(1) = \frac{\pi}{4}$$

$$\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$\text{mod}(z_1 \cdot z_2) = \text{mod}(z_1) \cdot \text{mod}(z_2)$$

$$|z_1 \cdot z_2| = |z_1| \cdot |z_2|$$

$$\text{mod}\left(\frac{z_1}{z_2}\right) = \frac{\text{mod}(z_1)}{\text{mod}(z_2)}$$

$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$$

$$\arg(z_1 \cdot z_2) = \arg(z_1) + \arg(z_2)$$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

the square root $x + iy$
الجزء الحقيقي x y الجزء التخيلي

$$\sqrt{x + iy} = \pm \left(\sqrt{\frac{r+x}{2}} + \sqrt{\frac{r-x}{2}} i \right)$$

$$\sqrt{x - iy} = \pm \left(\sqrt{\frac{r+x}{2}} - \sqrt{\frac{r-x}{2}} i \right)$$

The polar form
الصورة القطبية

$$[r, \theta] = r(\cos(\theta) + i \sin(\theta))$$

$$[r, -\theta] = r(\cos(\theta) - i \sin(\theta))$$

$$\cos(-\theta) = \cos(\theta)$$

$$\sin(-\theta) = -\sin(\theta)$$

⊛ If $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$ (ملاحظة)

$$z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$$

then

$$(1) \quad z_1 \cdot z_2 = r_1 r_2 (\cos (\theta_1 + \theta_2) + i \sin (\theta_1 + \theta_2))$$

نضرب الأطوال ونجمع الزوايا

$$(2) \quad \frac{z_1}{z_2} = \frac{r_1}{r_2} (\cos (\theta_1 - \theta_2) + i \sin (\theta_1 - \theta_2))$$

نقسم الأطوال ونطرح الزوايا

$$(3) \text{ If } z = r (\cos \theta + i \sin \theta) \text{ then } \bar{z} = r (\cos \theta - i \sin \theta)$$

تغير إشارة الجزء التخيلي

ملاحظة

$$z + \bar{z} = 2r \cos \theta$$

$$z - \bar{z} = 2ir \sin \theta$$

$$\text{If } (r=1)$$

then

$$z + \bar{z} = 2 \cos \theta$$

$$z - \bar{z} = 2i \sin \theta$$

$$\sim \text{ If } z = \cos \theta + i \sin \theta$$

$$\text{then } z + \frac{1}{\bar{z}} = 2 \cos \theta$$

$$z - \frac{1}{\bar{z}} = 2i \sin \theta$$

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1) If $z_1 = \cos(\alpha) + i \sin(\alpha)$, $z_2 = \cos(\beta) + i \sin(\beta)$
 then argument of $(z_1 \cdot z_2) = \dots$

$$\begin{aligned} \arg(z_1 \cdot z_2) &= \alpha \cdot \beta & \arg(z_1 \cdot z_2) &= \alpha - \beta \\ \arg(z_1 \cdot z_2) &= \alpha + \beta & & \text{none of these} \end{aligned}$$

2) If $z_1 = \cos(\alpha) + i \sin(\alpha)$, $z_2 = \cos(\beta) + i \sin(\beta)$
 then argument of $\frac{z_1}{z_2}$ is $\dots$

$$\begin{aligned} \arg\left(\frac{z_1}{z_2}\right) &= \frac{\alpha}{\beta} & \arg\left(\frac{z_1}{z_2}\right) &= \alpha - \beta \\ \arg\left(\frac{z_1}{z_2}\right) &= \beta - \alpha & & \text{none of these} \end{aligned}$$

3) If $z_1 = \cos(\alpha) + i \sin(\alpha)$
 $z_2 = \cos(\beta) + i \sin(\beta)$

then $\arg\left(\frac{z_2}{z_1}\right) = \dots$

$$\begin{aligned} & \frac{\beta}{\alpha} & \beta + \alpha & \beta - \alpha & \frac{\alpha}{\beta} \end{aligned}$$

4) If $z = \cos(\theta) + i \sin(\theta)$, then $z + \bar{z} = \dots$

$$\begin{aligned} & 2 \cos(2\theta) & 2 \sin(\theta) & 2 \cos(\theta) & 2i \sin \theta \end{aligned}$$

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(5) If $z = \cos(\theta) + i \sin \theta$ then $z - \bar{z} = \dots$

$2 \sin(2\theta)$ $2i \sin(\theta)$ $-2i \sin(\theta)$ $2 \cos(\theta)$

(6) If $z = \cos(\theta) + i \sin \theta$
then $z + \frac{1}{z} = \dots$

$2 \cos(\theta)$ $2 \cos(2\theta)$ $2i \sin(\theta)$ none

(7) If $z = \cos(\theta) + i \sin(\theta)$
then $z - \frac{1}{z} = \dots$

$2i \sin(2\theta)$ $2 \sin(\theta)$ $2i \sin(\theta)$ $2 \cos(\theta)$

(8) $\frac{\cos(\theta) + i \sin(\theta)}{\cos(\theta) - i \sin(\theta)} = \dots$

$\cos(2\theta) + i \sin(2\theta)$ $\cos(2\theta) - i \sin(2\theta)$

none of the

(9) $\frac{\cos(3\theta) + i \sin(3\theta)}{\cos(2\theta) - i \sin(2\theta)} = \dots$

$\cos(5\theta) + i \sin(5\theta)$

none
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 $\sin(5\theta) + i \sin(5\theta)$